

Unit 4. Expectation, Variance and Higher Moments

Mean, median, Jensen's inequality, variance, skewness, kurtosis

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 11–12, CC BY-NC-SA 4.0.

Roadmap

- 1 Expected value
- 2 Median and quantiles
- 3 Jensen's inequality
- 4 Variance and standard deviation
- 5 Higher moments
- 6 Order statistics
- 7 Wrap-up

From a full distribution to a single number

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Caveat. The mean need not be a “typical” value: a fair die has $\mathbb{E}[X] = 3.5$, never observed.

Bernoulli and binomial via indicators

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Lesson

The **indicator trick** + linearity of expectation is the fastest way to compute means of complicated random variables. We will reuse it throughout the course.

Properties of expectation

Algebra

- ① $\mathbb{E}[c] = c.$
- ② $\mathbb{E}[aX + b] = a \mathbb{E}[X] + b.$
- ③ **Linearity:** $\mathbb{E}[\sum_i a_i X_i + b] = \sum_i a_i \mathbb{E}[X_i] + b$ (no independence required).
- ④ **Multiplicativity:** $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ (only if $X \perp Y$).

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LOTUS (Law of the Unconscious Statistician)

For any function g ,

$$\mathbb{E}[g(X)] = \int g(x) f_X(x) dx \quad (\text{no need to derive the distribution of } g(X)).$$

Another measure of centre: the median

Definition

$$\text{med}(X) = \min \left\{ m \in \mathbb{R} : \mathbb{P}(X \leq m) \geq \frac{1}{2} \right\}.$$

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Mean vs. median.

- For symmetric distributions, mean = median.
- For right-skewed distributions (e.g. income), mean > median.
- Median is robust to outliers; mean is not.

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- **Median commutes with monotone transforms.** $\text{med}(u(X)) = u(\text{med}(X))$ when u is strictly increasing.
- **Median is **not** linear.** $\text{med}(X + Y) \neq \text{med}(X) + \text{med}(Y)$ in general.
- **Quantiles of averages $\neq$ averages of quantiles.** Being above the 90th percentile in each component of a vector test does **not** put you above the 90th percentile in the total.

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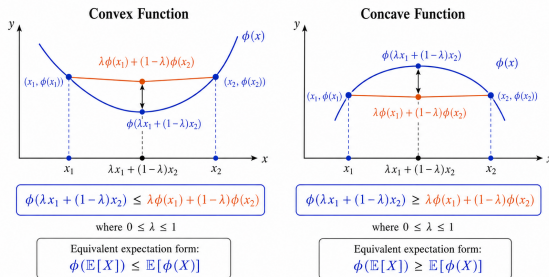
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Used everywhere

- Insurance and warranty pricing.
- Asset pricing (CAPM derives from Jensen-style reasoning).
- Behavioural / experimental economics.

Should you buy the warranty?

- Laptop \$1,200; with prob. $p = 0.10$, repair adds \$1,100.
- Warranty covers all repairs for a flat \$115.
- Initial wealth $W_0 = \$4,800$, utility $u(c) = \sqrt{4,800 - c}$.

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Decision: buy the plan, even though expected loss without it is only \$110. The \$5 gap is the **risk premium**.

Variance: how spread out is the distribution?

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Properties

- ① **Computational:** $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
- ② **Affine:** $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (no b inside).
- ③ **Independence:** if $X_1, \dots, X_n$ independent, $\text{Var}\left(\sum_i a_i X_i\right) = \sum_i a_i^2 \text{Var}(X_i)$.
- ④ $\text{Var}(X) = 0$ iff X is a constant.

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Two-line summary of the binomial

$$Y \sim B(n, p) \Rightarrow \mathbb{E}[Y] = np, \text{Var}(Y) = np(1 - p).$$

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Answer. $c = \mathbb{E}[X]$:

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Why this matters for econometrics

The variance is the **minimum mean-squared deviation** around any constant. This is the optimisation logic that, generalised to *conditional* mean-squared error, will give us the OLS regression in Unit 9 and the conditional expectation in Unit 6.

Skewness and kurtosis

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- $\gamma_1 > 0$: long right tail (income, firm size).
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Kurtosis: $\gamma_2 = \mu_4/\sigma^4$.

- Normal distribution has $\gamma_2 = 3$; **excess kurtosis** $\gamma_2 - 3$ is the standard reporting convention.
- Positive excess kurtosis = **fat tails**: extreme outcomes more likely than the normal would predict.
- This is the empirical norm for daily asset returns — and the basis for the criticism of Gaussian risk models in the 2008 crisis.

Order statistics of an i.i.d. sample

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Special cases.

- Maximum: $f_{Y_n}(y) = n F_X(y)^{n-1} f_X(y)$.
- Minimum: $f_{Y_1}(y) = n [1 - F_X(y)]^{n-1} f_X(y)$.

Auctions as order statistics

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- **Second-price (English) auction:** winner pays the second-highest bid; seller's revenue = Y_{n-1} , density $n(n-1)F_B(y)^{n-2}[1 - F_B(y)]f_B(y)$.

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Empirical auction theory — a major branch of industrial economics — relies on these order-statistic densities to estimate primitive bid distributions from observed prices.

Take-aways

- 1 Mean = centre of mass; median = centre by rank; they differ when distributions are skewed.
- 2 **Linearity** of expectation makes computation easy — no independence needed.
- 3 **Jensen's inequality**: $\mathbb{E}[u(X)] \geq u(\mathbb{E}[X])$ for convex u . Concave utility $\Rightarrow$ risk aversion.
- 4 Variance: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, additive across independent summands. Centring at the mean minimises mean-squared error — the OLS principle in disguise.
- 5 Skewness and kurtosis pick up asymmetry and tail fatness.

Next unit

Covariance, correlation, conditional expectation **and random vectors / multivariate framework.**

Thank you

Questions?

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